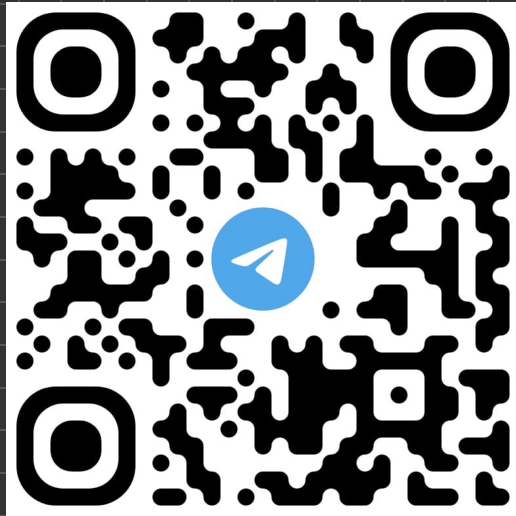


Telegram



Discord



video

logical

loop

bar bar repeat karna

for(initialization; condition; updation) {

log()

100 log()

→ atleast 1 bar

do {

} while(condition)

} while(condition)

}

function

Project → discount calculate

function keyword

function name

```
{  
function discount(p, d) {  
    return p * d / 100  
}
```

shirt → 5%
shoes → 10%

$$\frac{2500 \times 5}{100}$$

$$125$$

1000 → 0%

discount(2500, 5)

arguments

School

12x100

+ + +

add ()

function $\text{dis}(\text{al}(\underline{p}, d)) \{$

al $\rightarrow$ dis

return

calculation

undefined

input $\rightarrow$ function $\rightarrow$ new output

let discount = $\text{dis}(\text{al}(2r, 5))$

Frontend
Backend

⇒

clg X

debug

The diagram is drawn on a dark grid background. On the left, the words 'Frontend' and 'Backend' are stacked vertically. A large, light-colored curved arrow starts from the left of 'Frontend', loops around the left side of both words, and points towards the right. In the center, there is a double-lined arrow pointing to the right. To the right of this arrow, the text 'clg X' is written. A large, light-colored 'X' is drawn over the 'clg X' text. Below the 'clg X' text, there is a bracket-like shape that points down to the word 'debug'.

final price (

) { gst, discount,
shipping

}

type x

return

→ number
0 → Data type

void

undefined

let data =

0

data

+, -, *, /
↓
Math

→ Built in objects → kal

() banu banaye function

→ -5, → positive -1

```
✓  
function convertToPositiveNumber(num){  
  return num * -1;  
}  
let positiveNum = convertToPositiveNumber(-10)  
  
console.log(positiveNum);
```

} custom implementation

```
let positiveNum2 = Math.abs(-10)  
console.log(positiveNum2);
```

↗ object

↘ built in

↳ abs
absolute

Math ↗ object
↳ abs()
↳ function

Math. —
—
—
—
—
—

The JavaScript **Math** object is a built-in, global object used to perform mathematical operations on numbers.  W3Schools +1

5.4
6
5

5.6
6

round
decimal
delete

4.0 ✓
4

4.1

random()

↓

include → 0 → 1 → exclude

include → [0, 1) → exclude

0 * 2

↳ 0

$$8-3$$

$$\rightarrow 5 + 1$$

Range

$$\min = 3$$

$$\max = 6$$

include

$$\downarrow$$

$$50 - 100$$

$$1 - 6$$

$$0$$

$$\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 3 & 4 & 5 & 6 & 7 & 8 \end{array}$$

$$0 \rightarrow 10 \rightarrow 189963 \quad 6$$

$$(\text{Math.random}) * (\max - \min + 1)$$

$$\text{floor} \left(\frac{5.93937}{6} \right) + \min$$

$$6 + \min$$

$$6 + 3 \rightarrow 9$$

$$5 + 3$$

$$\rightarrow 8$$

$50 - 100$
 $\rightarrow 50$
 $8 - 3$
 $5 + 1$
 (6)

$3 \leftrightarrow 8$
 $\min = 3 \checkmark$
 $\max = 8 \checkmark$
 $\{0, 1\}$

$\rightarrow 3 \ 4 \ 5 \ 6 \ 7 \ 8$
 $\underline{\underline{1 \ 2 \ 3 \ 4 \ 5 \ 6}}$

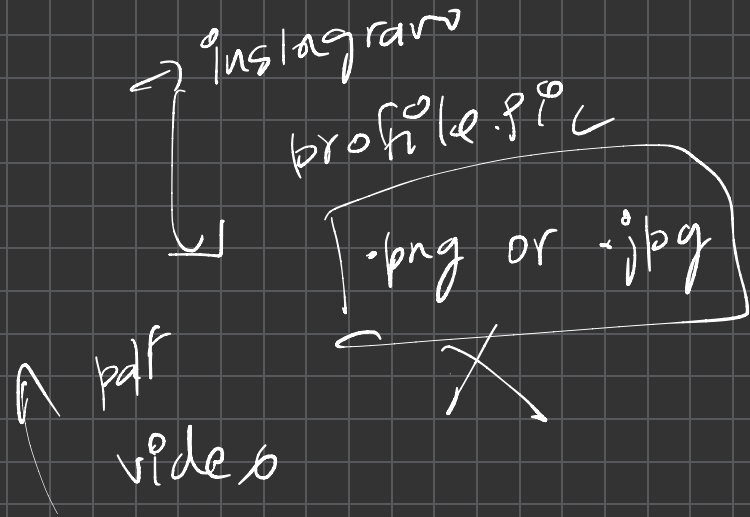
$\text{floor} (\text{Math.Random}() * (\max - \min + 1)) + \min$

$\downarrow$
 $\min \rightarrow 0$
 $\rightarrow \min + 0$
 $3 + 0$
 $\rightarrow 3$

$\downarrow$
 $\min = 5$
 $\rightarrow 5 + 3$
 $\neq 8$

Number

String



Date

1 January, 1970

unix timestamp

1787329266

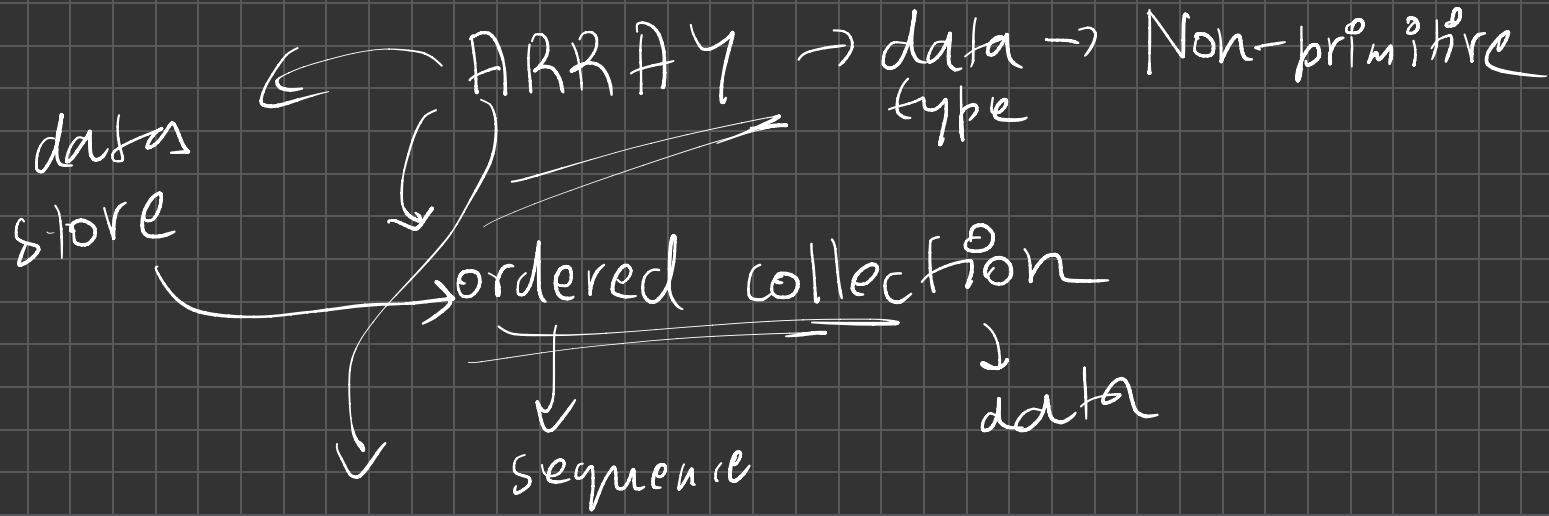
SECONDS SINCE JAN 01 1970. (UTC)

9:51:18 PM

Sun Mon Tue Wed Thu Fri Sat
1 2 3 4 5 6 7
0 1 2 3 4 5 6

Jan → 0

Dec → 11



`let kuchBhi = []`

variable

square brackets
array

array syntax

→
()
{ } →
[] →

`["tshirt", "lower", "cap", "shoes"]`

`let products = ["tshirt", "lower", "cap"]`

Indexing → array

memory offset ✓

products =

["tshirt", "lower", "cap", "shoes"]

↑
0

2

3

4 cly

↑
1

↑

2

↑
3

index
products[0]

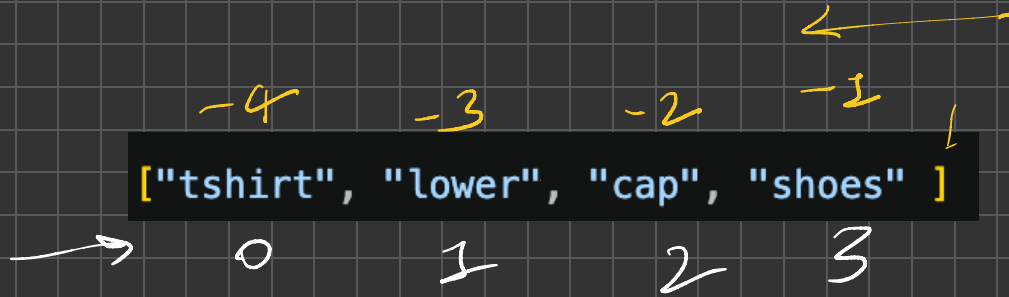
↑
MATLAB
FORTRAN

Length of Array

$\Sigma ["a", "b", "c", "d"]$
0 1 2 ③ $\rightarrow$
length = 4

length = total index + 1

$\boxed{\text{length} - 1} = \text{total index} \rightarrow \text{last index}$
 $\rightarrow \text{array}[\text{index}]$
 $\downarrow$ last index



for (`for(let i = 0 ; i < products.length; i++){ console.log(products[i]);` , `i < array.length ; i++`) {
 3

number of total items in array

```
for(let i = 0 ; i < products.length; i++){  
  console.log(products[i]);  
}
```

products[0]

variable index

i = 0
i = 1

Nested array

```
let products = ["tshirt", 566], ["lower", 543], ["cap", 53], ["shoes", 673]
```

products[0]

↳ ["tshirt", 566]

products[0][0]

product 1 product 2

```
let products = [{"tshirt", 566}, {"lower", 543}, {"cap", 53}, {"shoes", 673}]
```

0 1 2 3

```
let firstProduct = products[0] ✓  
console.log(firstProduct[0]);
```

```
console.log(products[0][0]);
```

firstprod[i]

```
let products = ["tshirt", "lower", "cap", "shoes"]
```

```
products.push("hello") // last mai add krta hai
```

```
products.pop() // last se delete karo
```

```
products.unshift("hello") // start mai add krta hai
```

```
products.shift() // start se delete krta hai
```

```
console.log(products);
```